CS23A34

USER INTERFACE DESIGN

EXPERIMENT-3

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1.COMMAND LINE INTERFACE

CLI Python Code:

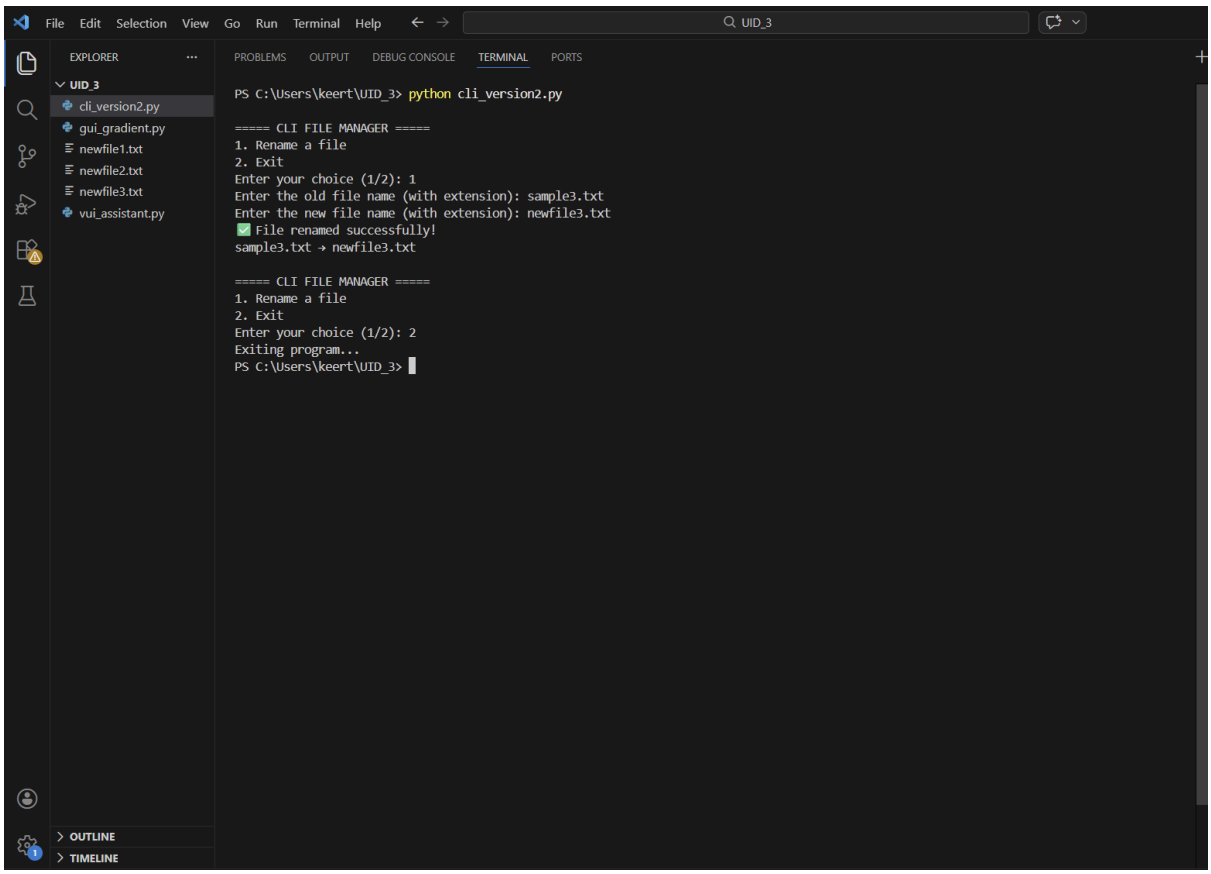
cli_app.py:

```
import os
def rename_file():
   old_name = input("Enter the old file name (with extension): ")
   new_name = input("Enter the new file name (with extension): ")
    try:
      if not os.path.isfile(old_name):
         print("❌ File does not exist.")
         return
   if os.path.isfile(new_name):
         print("⚠ A file with new name already exists.")
         return
```

```
        os.rename(old_name, new_name)
        print("✅ File renamed successfully!")
        print(f"{old_name} → {new_name}")

    except Exception as e:
        print("⚠️ Error occurred:", e)

def main():
    while True:
        print("\n===== CLI FILE MANAGER =====")
        print("1. Rename a file")
        print("2. Exit")

        choice = input("Enter your choice (1/2): ")

        if choice == "1":
            rename_file()
        elif choice == "2":
            print("Exiting program...")
            break
        else:
            print("❌ Invalid choice. Try again.")

if __name__ == "__main__":
```

```
main()
```

OUTPUT :



GUI (PYTHON CODE) :

```
import tkinter as tk
from tkinter import messagebox
import os


def rename_file():
    old_name = old_entry.get()
    new_name = new_entry.get()
```

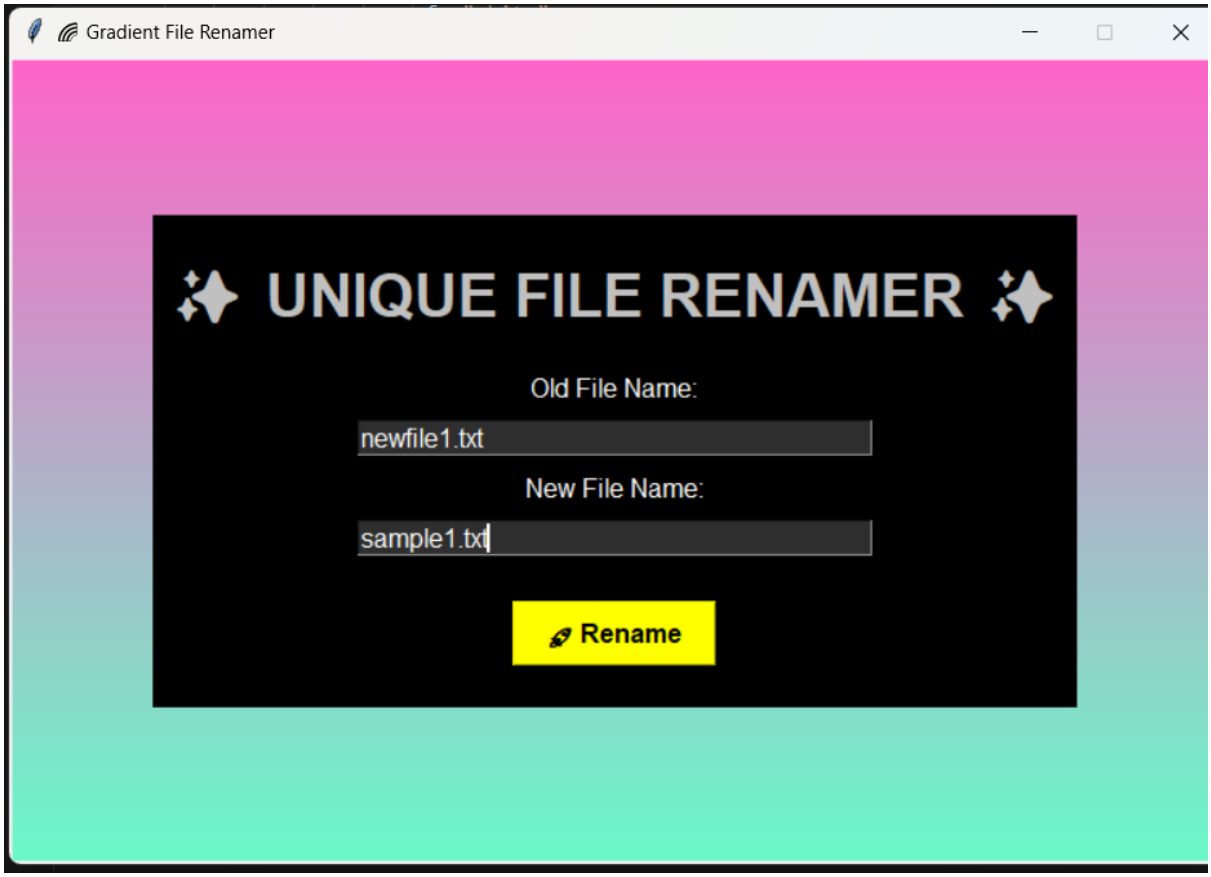
```
    if old_name == "" or new_name == "":
        messagebox.showwarning("Input Error", "Please enter both file
names!")
        return

    if not os.path.isfile(old_name):
        messagebox.showerror("Error", "File not found!")
        return

    if os.path.isfile(new_name):
        messagebox.showwarning("Warning", "New file name already exists!")
        return

    os.rename(old_name, new_name)
    messagebox.showinfo("Success 🎉", f"{old_name} ➜ {new_name}")
    old_entry.delete(0, tk.END)
    new_entry.delete(0, tk.END)

# Create main window
root = tk.Tk()
root.title("🌈 Gradient File Renamer")
root.geometry("750x500")
root.resizable(False, False)
```

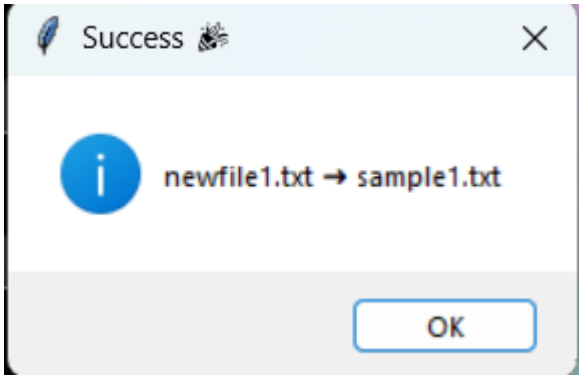
```
canvas = tk.Canvas(root, width=750, height=500)
canvas.pack(fill="both", expand=True)

# SAFE GRADIENT (Values always 0–255)
for i in range(500):
    r = max(0, min(255, int(255 - i * 0.3)))
    g = max(0, min(255, int(100 + i * 0.3)))
    b = 200
    color = f'#{r:02x}{g:02x}{b:02x}'
    canvas.create_line(0, i, 750, i, fill=color)

frame = tk.Frame(root, bg="black", bd=5)
frame.place(relx=0.5, rely=0.5, anchor="center")

tk.Label(frame,
        text="✨ UNIQUE FILE RENAMER ✨",
        font=("Helvetica", 28, "bold"),
        bg="black",
        fg="#c0c0c0").pack(pady=20)

tk.Label(frame,
        text="Old File Name:",
        font=("Arial", 12),
        bg="black",
        fg="white").pack()
```

```
old_entry = tk.Entry(frame,
                     width=35,
                     font=("Arial", 12),
                     bg="#2e2e2e",
                     fg="white",
                     insertbackground="white")
old_entry.pack(pady=8)

tk.Label(frame,
         text="New File Name:",
         font=("Arial", 12),
         bg="black",
         fg="white").pack()

new_entry = tk.Entry(frame,
                     width=35,
                     font=("Arial", 12),
                     bg="#2e2e2e",
                     fg="white",
                     insertbackground="white")
new_entry.pack(pady=8)

tk.Button(frame,
          text="🚀 Rename",
```

```
        font=("Arial", 12, "bold"),
        bg="yellow",
        fg="black",
        activebackground="orange",
        padx=15,
        pady=5,
        command=rename_file).pack(pady=20)

root.mainloop()
```

OUTPUT :





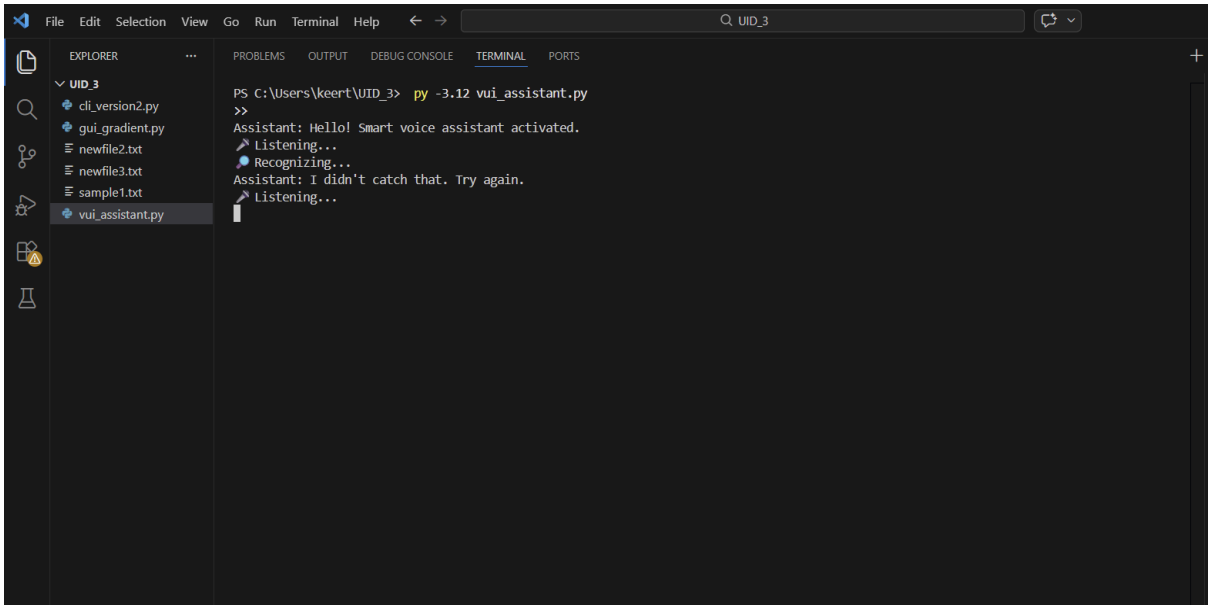
VUI (PYTHON CODE):

```
import speech_recognition as sr
import pyttsx3
import datetime
import webbrowser

# Initialize recognizer & speaker
recognizer = sr.Recognizer()
engine = pyttsx3.init()

def speak(text):
    print("Assistant:", text)
    engine.say(text)
    engine.runAndWait()

def listen():
```

```
    with sr.Microphone() as source:
        print(" 🎤 Listening...")
        recognizer.adjust_for_ambient_noise(source, duration=1)
        audio = recognizer.listen(source)

    try:
        print(" 🔎 Recognizing...")
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        return command.lower()
    except:
        speak("Sorry, I didn't understand.")
        return None

def main():
    speak("Hello! I am your voice assistant.")

    while True:
        command = listen()

        if command is None:
            continue
```

```
        if "hello" in command:
            speak("Hello boss!")

        elif "time" in command:
            current_time = datetime.datetime.now().strftime("%I:%M
%p")
            speak(f"The time is {current_time}")

        elif "open youtube" in command:
            speak("Opening YouTube")
            webbrowser.open("https://youtube.com")

        elif "exit" in command or "stop" in command:
            speak("Goodbye!")
            break

if __name__ == "__main__":
    main()
```

OUTPUT :


File Edit Selection View Go Run Terminal Help
UID_3
EXPLORER
UID_3
cli_version2.py
gui_gradient.py
newfile2.txt
newfile3.txt
sample1.txt
vui_assistant.py
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\keert\UID_3> py -3.12 vui_assistant.py
>>
Assistant: Hello! Smart voice assistant activated.
Listening...
Recognizing...
Assistant: I didn't catch that. Try again.
Listening...